THE GREATEST COMMON DIVISOR OF $\binom{n}{2},\dots,\binom{n}{n-2}$ EQUALS $e^{\Lambda(n)}e^{\Lambda(n-1)}$: A FORMAL PROOF OF MAIGA'S FORMULA FOR OEIS A328202

MATH0

Abstract. For $n \geq 4$ let $a(n) = \gcd\{\binom{n}{k} : 2 \leq k \leq n-2\}$, the greatest common divisor of row n of Pascal's triangle with the entries 1 and n removed (OEIS A328202). We prove that $a(n) = L(n)\,L(n-1)$ for every $n \geq 4$, where $L(m) = e^{\Lambda(m)}$ equals p if m is a power of the prime p and 1 otherwise (OEIS A014963); this identity was recorded in the OEIS as a conjecture by Jon Maiga in 2019. The result is not new: it follows from Theorems 4 and 7 of Kaplan and Levy (2004), whose paper predates the conjecture and is not cited in the OEIS entry, which still labels the formula a conjecture. We give a short self-contained proof, based on Kummer's theorem for $\binom{p^t}{k}$ and Lucas's congruence, and formalize it; to our knowledge this is the first formal proof of the result. The proof is verified in Lean 4 with Mathlib against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the listed terms of A328202 and A014963.

1. Introduction

The sequence A328202 of the OEIS [9], contributed by Joel Kaufmann on October 7, 2019, has offset 4 and is described as "the greatest common divisor of all the numbers in row n of Pascal's triangle excluding 1 and n". For example, row 8 is $1, 8, 28, 56, 70, 56, 28, 8, 1$, and $\gcd(28, 56, 70) = 14$, so $a(8) = 14$; the data section lists $a(4), \dots, a(78)$, beginning $6, 10, 5, 7, 14, 6, 3, 11, 11, 13, 13, 1, 2, 34$. On October 8, 2019, Jon Maiga added to the entry the formula

$$a(n) = \mathrm{A014963}(n) \cdot \mathrm{A014963}(n-1), \tag{1}$$

labelled as a conjecture, where A014963 [8] is the exponential of the von Mangoldt function: $\mathrm{A014963}(m) = e^{\Lambda(m)}$ is the prime p if m is a power p^t ($t \geq 1$) of p, and 1 otherwise. For $n = 8$ this reads $14 = 2 \cdot 7$. The entry also contains comments of Robert Israel (October 16, 2019): $p \mid a(p^k)$ and $p \mid a(p^k + 1)$ for every prime p and $k \geq 1$; $a(p) = 2p$ for a Fermat prime $p > 3$ and $a(p+1) = 2p$ for a Mersenne prime p; and the conjecture that these are the only cases with $a(n) > n$. A comment of Chai Wah Wu (October 9, 2026) notes that $\gcd(n, a(n)) = 1$ if n is not a prime power, by Ram's theorem. In the current revision of the entry (#46, October 10, 2026), (1) is still labelled "(conjecture)".

We prove (1) for every $n \geq 4$ (Theorem 2), with a proof verified in Lean 4.

Prior work and status. The identity (1) is a known theorem. We credit the literature as follows.

- Ram [10] determined in 1909 the greatest common divisor of the full interior row, $\gcd\{\binom{n}{k} : 1 \leq k \leq n-1\} = L(n)$ for $n \geq 2$; the comment of Chai Wah Wu in A328202 refers to this result as Ram's theorem.
- Kaplan and Levy [3] study $g(n,t) = \gcd\big(\binom{n}{t}, \binom{n}{t+1}, \dots, \binom{n}{n-t}\big)$, so that $a(n) = g(n,2)$. Their Theorem 4 is the formula $g(n,1) = L(n)$ for $n \geq 2$. They set $h(n,t) = g(n,t+1)/\operatorname{lcm}(g(n-1,t), g(n,t))$ for $n \geq 2$ and $0 \leq t < \lfloor n/2 \rfloor$ (their Notation 6, an integer by

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their Theorem 5), and their main result, Theorem 7, applied with $t = 1$ gives $h(n, 1) = 1$ for all $n \geq 4$ (the integer 1 has no prime factors). Hence

$$a(n) = g(n, 2) = \operatorname{lcm}\big(g(n-1, 1), g(n, 1)\big) = \operatorname{lcm}\big(L(n-1), L(n)\big) = L(n)L(n-1) \qquad (n \geq 4),$$

the last step because $L(n) \mid n$ and $L(n-1) \mid n-1$ are coprime. The same follows from their Theorem 9 with $t = 1$. Their Table 2 lists $g(n, 2) = 6, 10, 5, 7, 14, 6$ for $n = 4, \ldots, 9$, in agreement with A328202. Their proofs use Kummer's theorem on carries [4].

- Joris, Oestreicher and Steinig [2] study the greatest common divisor of the binomial coefficients $\binom{n}{k}$ with k in an interval $r \leq k \leq s$; according to later work citing it, they give an explicit general formula for this gcd, which would contain (1) as the case $r = 2$, $s = n-2$. We have not been able to consult this paper and have not checked its formula, so we do not know whether it states (1) explicitly.

The paper [3] appeared fifteen years before the conjecture was recorded, and neither [3] nor [2] is cited in the OEIS entry. The math0 referees found no formalization of the result in Lean or Mathlib. We therefore classify the present work as the *first formal proof of a known result*.

Our proof (Section 3) works one prime at a time and avoids base-p digits: it uses the p-adic valuation of $\binom{p^t}{k}$, the congruence $\binom{p^t m}{p^t} \equiv m \pmod p$ (from Lucas's theorem [5]), Pascal's rule and $(N+1)\binom{N}{j} = (j+1)\binom{N+1}{j+1}$. Israel's conjecture follows by a short argument that we did not formalize (Remark 7).

2. Statement

Definition 1. For $m \geq 1$ let $L(m) = p$ if $m = p^t$ for a prime p and an integer $t \geq 1$, and $L(m) = 1$ otherwise; thus $L(m) = e^{\Lambda(m)}$, where Λ is the von Mangoldt function. For $n \geq 4$ let $a(n) = \gcd\{\binom{n}{k} : 2 \leq k \leq n-2\}$.

For $n \geq 4$ the entries of row n equal to 1 or n are exactly $\binom{n}{0} = \binom{n}{n} = 1$ and $\binom{n}{1} = \binom{n}{n-1} = n$, since $\binom{n}{k} \geq \binom{n}{2} = n(n-1)/2 > n$ for $2 \leq k \leq n-2$. So $a(n)$ is the sequence A328202. The FORMULA section of the entry [9] contains, verbatim:

```
a(n) = A014963(n) * A014963(n - 1) (conjecture).
- _Jon Maiga_, Oct 08 2019
```

The name of A014963 [8] (offset 1) is "Exponential of Mangoldt function M(n): a(n) = 1 unless n is a prime or prime power, in which case a(n) = that prime."

Theorem 2. *For every integer $n \geq 4$,*

$$\gcd\left\{\binom{n}{k} : 2 \leq k \leq n-2\right\} = L(n)\,L(n-1).$$

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A328202.lean`, reviewed for fidelity and frozen (SHA-256 `4021583e...ca08f`) before any proof attempt. Its definitions and main statement read, verbatim (import, comments, docstrings, blank lines, `set_option` lines, all theorems and the secondary definition `statementIsrael` omitted):

```
namespace Math0.Statements.A328202
def a (n : ℕ) : ℕ := (Finset.Icc 2 (n - 2)).gcd (Nat.choose n)
def expMangoldt (m : ℕ) : ℕ := if IsPrimePow m then m.minFac else 1
def statement : Prop :=
  ∀ n : ℕ, 4 ≤ n → a n = expMangoldt n * expMangoldt (n - 1)
end Math0.Statements.A328202
```

Here `(Finset.Icc 2 (n - 2)).gcd (Nat.choose n)` is the greatest common divisor of $\binom{n}{k}$ over $2 \le k \le n-2$, which is $a(n)$ by the discussion above. (The programs in the OEIS entry take the gcd over $2 \le k \le n/2$, or $2 \le k \le \lfloor (n+1)/2 \rfloor$ in the PARI program; by the symmetry $\binom{n}{k} = \binom{n}{n-k}$ this is the same set of numbers.) Mathlib's `IsPrimePowm` means $m = p^t$ with p prime and $t \ge 1$, and then `m.minFac`, the least prime factor of m, is p; so `expMangoldt` is L. Since $n \ge 4$, the truncated subtractions $n-1$ and $n-2$ are exact, the range $2 \le k \le n-2$ is nonempty, and the statement covers the whole range $n \ge 4$ of the sequence with no extra hypotheses. The values of `a` for $n < 4$ play no role.

The statement file also proves three facts that tie the definitions to their sources:

```
theorem a_data : (List.range 75).map (fun i => a (i + 4)) = [6, 10, 5, 7, 14, ...]
theorem expMangoldt_data : (List.range 95).map (fun i => expMangoldt (i + 1)) = [...]
theorem cast_expMangoldt (m : ℕ) :
    (expMangoldt m : ℝ) = Real.exp (ArithmeticFunction.vonMangoldt m)
```

In `a_data` the list (elided here) is the complete data section $a(4), \dots, a(78)$ of A328202, and `expMangoldt_data` compares `expMangoldt` with all 95 terms listed in A014963, for $m = 1, \dots, 95$. Both are proved by `decide +kernel`, that is, by evaluating the definitions in the Lean kernel (about four seconds for `a_data`). The lemma `cast_expMangoldt` identifies `expMangoldt` with e^{Λ} for Mathlib's von Mangoldt function. The file also records Israel's conjecture as a separate proposition `statementIsrael`; it is not the target of this work and is not proved in Lean.

3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A328202.lean`; the Lean name of each step is given in brackets. Throughout, p denotes a prime and v_p the p-adic valuation. We use two classical facts from Mathlib: the identity

$$(N+1)\binom{N}{j} = \binom{N+1}{j+1}(j+1) \qquad (N, j \ge 0) \tag{2}$$

(`Nat.add_one_mul_choose_eq`), and Pascal's rule $\binom{N+1}{j+1} = \binom{N}{j} + \binom{N}{j+1}$.

Lemma 3 (Kummer for prime powers; `not_sq_dvd_choose`)**.** *Let $t \ge 0$ and $1 \le k \le p^t$. Then $v_p\binom{p^t}{k} = t - v_p(k)$. In particular, if $v_p(k) \ge t-1$, then $p^2 \nmid \binom{p^t}{k}$.*

Proof. Write $N = p^t$. Then

$$\binom{N}{k} = \frac{N(N-1)\cdots(N-k+1)}{k!} = \frac{N}{k}\prod_{j=1}^{k-1}\frac{N-j}{j}.$$

For $1 \le j \le k-1 < p^t$ put $v = v_p(j)$; then $v < t$, so $p^v \mid p^t - j$, while $p^{v+1} \mid p^t$ and $p^{v+1} \nmid j$ give $p^{v+1} \nmid p^t - j$. Hence $v_p(N-j) = v_p(j)$, every factor of the product has valuation 0, and $v_p\binom{N}{k} = t - v_p(k)$. If $v_p(k) \ge t-1$ this is at most 1. (In Lean the formula is Mathlib's `Nat.factorization_choose_prime_pow`.) □

Lemma 4 (`expMangoldt_dvd_self`, `expMangoldt_prime_pow`)**.** *For $m \ge 1$, $L(m) \mid m$; and $L(p^t) = p$ for $t \ge 1$. Consequently, for $n \ge 2$, $L(n)$ and $L(n-1)$ are coprime.*

Proof. The first two assertions are immediate from the definition. Since $\gcd(n, n-1) = 1$, any divisors of n and of $n-1$ are coprime. □

Lemma 5 (`expMangoldt_dvd_choose`, `expMangoldt_pred_dvd_choose`)**.** *Let $n \ge 1$.*

(1) *$L(n) \mid \binom{n}{k}$ for $0 < k < n$.*

(2) $L(n-1) \mid \binom{n}{k}$ *for* $2 \le k \le n-2$.

Proof. (1) If $L(n) = 1$ there is nothing to prove. Otherwise $n = p^t$ with $t \ge 1$ and $L(n) = p$; since $0 < k < p^t$ we have $v_p(k) < t$, so $v_p\binom{p^t}{k} = t - v_p(k) \ge 1$ by Lemma 3. (In Lean this is Mathlib's `Nat.Prime.dvd_choose_pow`.)

(2) Write $n = N + 1$ and $k = j + 1$, so that $1 \le j$ and $j + 1 \le N - 1$. By Pascal's rule $\binom{n}{k} = \binom{N}{j} + \binom{N}{j+1}$ with $0 < j < j + 1 < N$, and by (1) applied to $N = n - 1$ both terms are divisible by $L(n-1)$. □

Lemma 6 (Lucas; `choose_pow_mul_modEq`, `not_dvd_choose_pow_mul`)**.** *For all* $m, t \ge 0$,

$$\binom{p^t m}{p^t} \equiv m \pmod{p}.$$

In particular, if $p \nmid m$ *then* $p \nmid \binom{p^t m}{p^t}$.

Proof. We use one step of Lucas's theorem [5] (Mathlib's `Choose.choose_modEq_choose_mod_mul_choose_div_nat`): for all $N, K \ge 0$,

$$\binom{N}{K} \equiv \binom{N \bmod p}{K \bmod p}\binom{\lfloor N/p \rfloor}{\lfloor K/p \rfloor} \pmod{p}. \tag{3}$$

(Proof: write $N = N_0 + pN'$ and $K = K_0 + pK'$ with $0 \le N_0, K_0 < p$. In $\mathbb{F}_p[x]$ we have $(1+x)^N = (1+x)^{N_0}(1+x^p)^{N'}$, and the coefficient of x^K on the right is $\sum \binom{N_0}{i}\binom{N'}{j}$ over $i + pj = K$, $0 \le i \le N_0$; since $i \equiv K \pmod{p}$ and $0 \le i < p$, the only possible term is $i = K_0$, $j = K'$, and if $K_0 > N_0$ there is no term at all, matching $\binom{N_0}{K_0} = 0$.) We prove the lemma by induction on t. For $t = 0$, $\binom{m}{1} = m$. For the step, apply (3) with $N = p^{t+1}m$ and $K = p^{t+1}$: here $N \bmod p = K \bmod p = 0$, $\lfloor N/p \rfloor = p^t m$ and $\lfloor K/p \rfloor = p^t$, so $\binom{p^{t+1}m}{p^{t+1}} \equiv \binom{0}{0}\binom{p^t m}{p^t} \equiv m \pmod{p}$ by the induction hypothesis. □

Proof of Theorem 2 (`result`). Fix $n \ge 4$ and put $g = a(n)$ and $\ell = L(n)L(n-1)$. By definition of the gcd,

$$g \mid \binom{n}{k} \qquad \text{for all } 2 \le k \le n-2. \tag{4}$$

We show $\ell \mid g$ and $g \mid \ell$; then $g = \ell$.

Step 1: $\ell \mid g$. Let $2 \le k \le n-2$. By Lemma 5, $L(n) \mid \binom{n}{k}$ and $L(n-1) \mid \binom{n}{k}$, and these two numbers are coprime by Lemma 4, so $\ell \mid \binom{n}{k}$. As this holds for every k in the range, ℓ divides the gcd g.

Step 2: $g \mid \ell$. It suffices to show that $p^e \mid g$ implies $p^e \mid \ell$ for every prime p and every $e \ge 0$ (Mathlib's `Nat.dvd_iff_prime_pow_dvd_dvd`). For $e = 0$ this is trivial, so let $e \ge 1$ and $p^e \mid g$; in particular $p \mid g$. We record two observations.

(i) If $p \mid \ell$ and $p^2 \nmid g$, then $p^e \mid \ell$. Indeed $p^2 \nmid g$ forces $e \le 1$, so $p^e \mid p \mid \ell$.

(ii) $p \mid n$ or $p \mid n-1$. Indeed, (2) with $N = n-1$, $j = 1$ reads $n(n-1) = 2\binom{n}{2}$, and $p \mid g \mid \binom{n}{2}$ by (4).

Case A: $p \mid n$. Write $n = p^t m$ with $p \nmid m$; then $t \ge 1$ (because $p \mid n$) and $m \ge 1$.

Case A1: $m = 1$, i.e. $n = p^t$. Then $L(n) = p$ by Lemma 4, so $p \mid \ell$, and by (i) it suffices to show $p^2 \nmid g$. We exhibit k_0 with $2 \le k_0 \le n-2$ and $p^2 \nmid \binom{n}{k_0}$; then $p^2 \nmid g$ by (4). If $t = 1$, take $k_0 = 2$: it lies in the range because $n \ge 4$, and $2 \le p = p^t$ and $v_p(2) \ge 0 = t - 1$, so $p^2 \nmid \binom{p}{2}$ by Lemma 3. If $t \ge 2$, take $k_0 = p^{t-1}$. Then $k_0 \ge p \ge 2$ and $k_0 + 2 \le 2k_0 \le pk_0 = n$, so k_0 lies in the range; moreover $v_p(k_0) = t - 1$, so $p^2 \nmid \binom{p^t}{k_0}$ by Lemma 3.

Case A2: $m \geq 2$. This case is impossible. Take $k_0 = p^t$. Then $k_0 \geq p \geq 2$ and $k_0 + 2 \leq 2p^t \leq mp^t = n$, so $p \mid g \mid \binom{n}{p^t} = \binom{p^t m}{p^t}$ by (4), contradicting Lemma 6.

Case B: $p \mid n-1$. Write $n - 1 = p^t m$ with $p \nmid m$; again $t \geq 1$ and $m \geq 1$. Note that $p \nmid p^t m + 1 = n$, since $p \mid p^t m$.

Case B1: $m = 1$, i.e. $n - 1 = p^t$. Then $L(n-1) = p$, so $p \mid \ell$, and by (i) it suffices to show $p^2 \nmid g$. Take $k_0 = p^{t-1} + 1 \geq 2$. It lies in the range: if $t = 1$ then $k_0 = 2$ and $k_0 + 2 = 4 \leq n$; if $t \geq 2$ then $p^{t-1} \geq 2$ and $p^t \geq 2p^{t-1}$, so $k_0 + 2 = p^{t-1} + 3 \leq 2p^{t-1} + 1 \leq p^t + 1 = n$. By (2) with $N = p^t$ and $j = p^{t-1}$, we have $(p^t + 1)\binom{p^t}{p^{t-1}} = \binom{n}{k_0}(p^{t-1} + 1)$. Suppose $p^2 \mid g$. Then $p^2 \mid \binom{n}{k_0}$ by (4), so p^2 divides the left-hand side. As $p \nmid p^t + 1$, the numbers p^2 and $p^t + 1$ are coprime, hence $p^2 \mid \binom{p^t}{p^{t-1}}$. This contradicts Lemma 3, since $v_p(p^{t-1}) = t - 1$. Hence $p^2 \nmid g$.

Case B2: $m \geq 2$. This case is impossible. Take $k_0 = p^t + 1 \geq 2$; then $k_0 + 2 = p^t + 3 \leq 2p^t + 1 \leq mp^t + 1 = n$, because $p^t \geq 2$. By (2) with $N = p^t m$ and $j = p^t$, we have $(p^t m + 1)\binom{p^t m}{p^t} = \binom{n}{k_0}(p^t + 1)$. Since $p \mid g \mid \binom{n}{k_0}$, the prime p divides the left-hand side, hence divides $p^t m + 1$ or $\binom{p^t m}{p^t}$. The first is impossible because $p \mid p^t m$, and the second contradicts Lemma 6.

By (ii), Cases A and B are exhaustive, and in every case that can occur we have shown $p^e \mid \ell$. Hence $g \mid \ell$, and with Step 1, $g = \ell$. □

Remark 7 (Israel's conjecture; not formalized)**.** Theorem 2 implies the conjecture of Israel recorded in A328202 (the "only if" direction below), together with the two facts he states before it (the "if" direction): for $n \geq 4$, $a(n) > n$ holds if and only if n is a Fermat prime > 3 or $n - 1$ is a Mersenne prime. Indeed, if at most one of n, $n - 1$ is a prime power, then $a(n)$ equals 1, $L(n) \leq n$ or $L(n-1) \leq n - 1$, so $a(n) \leq n$. If both are prime powers, one of them is even, hence a power of 2, and $a(n) = 2p$ where the other one is p^s with p an odd prime. If $s \geq 2$ then $2p < p^2 \leq p^s \leq n$, so $a(n) \leq n$. If $s = 1$, then either $n = p = 2^r + 1$ with $r \geq 2$, which is a Fermat prime > 3 (a prime of the form $2^r + 1$ has r a power of 2), or $n - 1 = p = 2^r - 1$ is a Mersenne prime; in both cases $a(n) = 2p > n$. Conversely, if n is a Fermat prime > 3 or $n - 1$ is a Mersenne prime, then n and $n - 1$ are both prime powers, the odd one is a prime p, and $a(n) = 2p > n$. This deduction is not part of the Lean development, and the proposition `statementIsrael` of the statement file is not proved there.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A328202.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A328202.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A328202.result`, of type `Math0.Statements.A328202.statement`. The auxiliary lemmas, all private, correspond to Lemmas 3–6; their statements read, verbatim (proofs and docstrings omitted):

```
private lemma expMangoldt_dvd_self (m : ℕ) : expMangoldt m ∣ m
private lemma expMangoldt_prime_pow {p t : ℕ} (hp : p.Prime) (ht : t ≠ 0) :
    expMangoldt (p ^ t) = p
private lemma expMangoldt_dvd_choose (n k : ℕ) (hk0 : k ≠ 0) (hkn : k ≠ n) :
    expMangoldt n ∣ n.choose k
private lemma expMangoldt_pred_dvd_choose (n k : ℕ) (hk : 2 ≤ k) (hkn : k + 2 ≤ n) :
    expMangoldt (n - 1) ∣ n.choose k
private lemma not_sq_dvd_choose {p t k : ℕ} (hp : p.Prime) (hk0 : k ≠ 0) (hk : k ≤ p ^ t)
    (hf : t - 1 ≤ k.factorization p) : ¬ p ^ 2 ∣ (p ^ t).choose k
private lemma choose_pow_mul_modEq {p m : ℕ} (hp : p.Prime) :
    ∀ t : ℕ, (p ^ t * m).choose (p ^ t) ≡ m [MOD p]
```

```
private lemma not_dvd_choose_pow_mul {p m t : ℕ} (hp : p.Prime) (hm : ¬ p ∣ m) :
    ¬ p ∣ (p ^ t * m).choose (p ^ t)
theorem result : statement
```

The proof of `result` follows Section 3 case by case; the decompositions $n = p^t m$ and $n-1 = p^t m$ come from `Nat.exists_eq_pow_mul_and_not_dvd`, and the range conditions on k_0 are closed by `omega` and `nlinarith`. The proof file contains no `decide`, `native_decide`, `sorry` or `set_option`.

Audit. The file ends with the command

```
#math0_settles Math0.Frontier.A328202.result Math0.Statements.A328202.statement
```

defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself. It fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (rejecting `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A328202.lean` in the directory `lean` prints

```
math0 audit: 'Math0.Statements.A328202.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A328202.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A328202.result' proves 'Math0.Statements.A328202.statement'
```

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use: the statement is a math0 formalization of an OEIS conjecture. To reproduce, run `scripts/verify.sh` from the repository root. It fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. Once its dependencies are built, the single file can also be checked with the `lake env lean` command above.

References

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[2] H. Joris, C. Oestreicher, and J. Steinig, *The greatest common divisor of certain sets of binomial coefficients*, J. Number Theory **21** (1985), no. 1, 101–119.

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[4] E. E. Kummer, *Über die Ergänzungssätze zu den allgemeinen Reciprocitätsgesetzen*, J. Reine Angew. Math. **44** (1852), 93–146.

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[6] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[7] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[8] The OEIS Foundation Inc., Entry A014963 in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A014963`.

[9] The OEIS Foundation Inc., Entry A328202 in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A328202` (revision #46, October 10, 2026).

[10] B. Ram, *Common factors of* $\frac{n!}{m!\,(n-m)!}$ $(m = 1, 2, \ldots, n-1)$, J. Indian Math. Club **1** (1909), 39–43.